



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT	AI-Driven Software Architecture and Codebase Intelligence	
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Abstract

Modern software systems evolve rapidly, and their architecture often drifts from its original design as developers make ad-hoc changes under deadline pressure. This project explores how AI and large language models (LLMs) can be used to continuously analyze, understand, and adapt software architecture as a codebase evolves, directly addressing the core challenge of adaptive software engineering: keeping systems architecturally sound as requirements and code change over time.

The system ingests a codebase and constructs a live architectural model by combining static analysis (dependency graphs, module coupling, layering violations) with LLM-based semantic understanding (inferring component responsibilities, detecting architectural intent from code and comments, summarizing design patterns in use). This hybrid approach, referred to as "codebase intelligence," allows the tool to answer questions a static analyser alone cannot, such as why a module exists or whether a new change violates the system's intended design, not just its explicit rules.

Building on this model, the project implements an adaptive feedback loop. As developers commit changes, the system detects architectural drift (for example, a service quietly gaining a dependency it shouldn't have), flags violations of established design principles, and suggests refactoring or documentation updates, closing the gap between how the system was designed and how it is actually evolving.

The expected outcome is a working prototype demonstrating automated architecture visualization, drift detection, and AI-generated refactoring suggestions on a sample multi-module codebase, along with an evaluation of how well AI-inferred architectural understanding matches ground-truth design intent.

